量子力学习题解答
第四章　态和力学量的表象

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教材：周世勋《量子力学教程》

正式解题之前，我们先做一些预备工作.

在引入了 Dirac 记号之后，我们有如下的单位分解.

$$\hat{I}=\sum_n |\psi_n\rangle\langle\psi_n|, \qquad \hat{I}=\int_\Omega \mathrm{d}\mathbf{r}\,|\psi_n\rangle\langle\psi_n|. \tag{1}$$

这里，ψ_n 是量子力学态空间的一组正交归一并且完备的基底，$\mathrm{d}\mathbf{r}$ 为全空间 Ω 的微元.

利用左矢和右矢的定义，我们可以将一个力学量 $\hat{O}$ 的矩阵元写作

$$\langle\alpha|\hat{O}|\beta\rangle=\langle\psi_\alpha|\hat{O}|\psi_\beta\rangle=(\psi_\alpha,\hat{O}\psi_\beta)=O_{\alpha,\beta}. \tag{2}$$

但是为了计算这一矩阵元，我们需要取一个具体的表象. 例如，我们可以取坐标表象，此时我们有

$$\langle\alpha|\hat{O}|\beta\rangle=\iint \mathrm{d}\mathbf{r}\mathrm{d}\mathbf{r}'\langle\psi_\alpha|\mathbf{r}'\rangle\langle\mathbf{r}'|\hat{O}|\mathbf{r}\rangle\langle\mathbf{r}|\psi_\beta\rangle=\iint \mathrm{d}\mathbf{r}\mathrm{d}\mathbf{r}'\psi_\alpha^*(\mathbf{r}')\hat{O}_{\mathbf{r}',\mathbf{r}}\psi_\beta(\mathbf{r}). \tag{3}$$

在上面的运算中，我们插入了两个单位分解. 当算符 $\hat{O}$ 仅与空间位置有关时，我们采用坐标表象为宜，此时我们有坐标表象中的坐标矩阵元

$$\hat{O}_{\mathbf{r}',\mathbf{r}}=\langle\mathbf{r}'|\hat{O}|\mathbf{r}\rangle=\hat{O}_{\mathbf{r}}\delta(\mathbf{r}'-\mathbf{r}). \tag{4}$$

我们也可以取动量表象，这时我们有

$$\langle\alpha|\hat{O}|\beta\rangle=\iint \mathrm{d}\mathbf{p}\mathrm{d}\mathbf{p}'\langle\psi_\alpha|\mathbf{p}'\rangle\langle\mathbf{p}'|\hat{O}|\mathbf{p}\rangle\langle\mathbf{p}|\psi_\beta\rangle=\iint \mathrm{d}\mathbf{p}\mathrm{d}\mathbf{p}'\psi_\alpha^*(\mathbf{p}')\hat{O}_{\mathbf{p}',\mathbf{p}}\psi_\beta(\mathbf{p}). \tag{5}$$

在上面的运算中，我们仍然插入了两个单位分解. 当算符 $\hat{O}$ 仅与体系动量有关时，我们采用动量表象为宜，此时我们有动量表象中的动量矩阵元：

$$\hat{O}_{\mathbf{p}',\mathbf{p}}=\langle\mathbf{p}'|\hat{O}|\mathbf{p}\rangle=\hat{O}_{\mathbf{p}}\delta(\mathbf{p}'-\mathbf{p}). \tag{6}$$

但有的时候，我们也需要计算动量表象中坐标的矩阵元 $x_{p',p}$、坐标表象中动量的矩阵元 $p_{x',x}$，我们以 $x_{p',p}$ 为例：

$$x_{p',p}=\langle p'|\hat{x}|p\rangle=\iint \mathrm{d}x'\mathrm{d}x\langle p'|x'\rangle\langle x'|\hat{x}|x\rangle\langle x|p\rangle=\iint \mathrm{d}x'\mathrm{d}x\langle p'|x'\rangle\langle x'|x|x\rangle\langle x|p\rangle$$

$$
\begin{aligned}
&= \iint \mathrm{d}x'\mathrm{d}x \langle p'\,|x'\rangle\, x \langle x'\,|x\rangle\, \langle x\,|p\rangle = \iint \mathrm{d}x'\mathrm{d}x \langle p'\,|x'\rangle\, x\delta(x'-x)\langle x\,|p\rangle \\
&= \iint \mathrm{d}x'\mathrm{d}x \left[\frac{1}{\sqrt{2\pi\hbar}}\exp\left(\frac{\mathrm{i}}{\hbar}p'x'\right)\right]^* x\delta(x'-x)\left[\frac{1}{\sqrt{2\pi\hbar}}\exp\left(\frac{\mathrm{i}}{\hbar}px\right)\right] \\
&= \frac{1}{2\pi\hbar}\int \mathrm{d}x\cdot x\exp\left[\frac{\mathrm{i}}{\hbar}(p-p')x\right] = \frac{1}{2\pi\hbar}\int \mathrm{d}x\cdot \frac{\hbar}{\mathrm{i}}\frac{\partial}{\partial p}\exp\left[\frac{\mathrm{i}}{\hbar}(p-p')x\right] \\
&= \frac{\hbar}{\mathrm{i}}\frac{\partial}{\partial p}\left(\frac{1}{2\pi\hbar}\int \mathrm{d}x\exp\left[\frac{\mathrm{i}}{\hbar}(p-p')x\right]\right) = \frac{\hbar}{\mathrm{i}}\frac{\partial}{\partial p}\delta(p-p'). \qquad (7)
\end{aligned}
$$

在上式的第一行，我们又一次插入了单位分解. 同理，我们可以证明

$$
p_{x',x} = \langle x'|\,\hat{p}\,|x\rangle = -\frac{\hbar}{\mathrm{i}}\frac{\partial}{\partial x}\delta(x-x'). \qquad (8)
$$

下面我们开始解题:

4.1

我们知道

$$
\hat{L}_x = \hat{y}\hat{p}_z - \hat{z}\hat{p}_y = \frac{\hbar}{\mathrm{i}}\left(y\frac{\partial}{\partial z} - z\frac{\partial}{\partial y}\right), \qquad \hat{L}_x^2 = -\hbar^2\left(y\frac{\partial}{\partial z} - z\frac{\partial}{\partial y}\right)^2. \qquad (9)
$$

欲求其在动量表象中的矩阵元，让我们仿照 (7) 式

1)

$$
\begin{aligned}
(\hat{L}_x)_{\mathbf{p}',\mathbf{p}} &= \langle \mathbf{p}'|\,\hat{L}_x\,|\mathbf{p}\rangle = \iint \mathrm{d}\mathbf{r}\mathrm{d}\mathbf{r}' \langle \mathbf{p}'\,|\mathbf{r}'\rangle\, \langle \mathbf{r}'|\,\hat{L}_x\,|\mathbf{r}\rangle\, \langle \mathbf{r}\,|\mathbf{p}\rangle \\
&= \iint \mathrm{d}\mathbf{r}\mathrm{d}\mathbf{r}' \left[\frac{1}{(2\pi\hbar)^{3/2}}\exp\left(\frac{\mathrm{i}}{\hbar}\mathbf{p}'\cdot\mathbf{r}'\right)\right]^* L_x\langle \mathbf{r}'\,|\mathbf{r}\rangle\, \frac{1}{(2\pi\hbar)^{3/2}}\exp\left(\frac{\mathrm{i}}{\hbar}\mathbf{p}\cdot\mathbf{r}\right) \\
&= \iint \mathrm{d}\mathbf{r}\mathrm{d}\mathbf{r}' \left[\frac{1}{(2\pi\hbar)^{3/2}}\exp\left(\frac{\mathrm{i}}{\hbar}\mathbf{p}'\cdot\mathbf{r}'\right)\right]^* L_x\delta(\mathbf{r}'-\mathbf{r})\frac{1}{(2\pi\hbar)^{3/2}}\exp\left(\frac{\mathrm{i}}{\hbar}\mathbf{p}\cdot\mathbf{r}\right) \\
&= \int \mathrm{d}\mathbf{r} \left[\frac{1}{(2\pi\hbar)^{3/2}}\exp\left(\frac{\mathrm{i}}{\hbar}\mathbf{p}'\cdot\mathbf{r}\right)\right]^* L_x\frac{1}{(2\pi\hbar)^{3/2}}\exp\left(\frac{\mathrm{i}}{\hbar}\mathbf{p}\cdot\mathbf{r}\right) \\
&= \int \mathrm{d}\mathbf{r}\frac{1}{(2\pi\hbar)^3}\exp\left(-\frac{\mathrm{i}}{\hbar}\mathbf{p}'\cdot\mathbf{r}\right) L_x\exp\left(\frac{\mathrm{i}}{\hbar}\mathbf{p}\cdot\mathbf{r}\right) \\
&= \int \mathrm{d}\mathbf{r}\frac{1}{(2\pi\hbar)^3}\exp\left(-\frac{\mathrm{i}}{\hbar}\mathbf{p}'\cdot\mathbf{r}\right)\frac{\hbar}{\mathrm{i}}\left(y\frac{\partial}{\partial z} - z\frac{\partial}{\partial y}\right)\exp\left(\frac{\mathrm{i}}{\hbar}\mathbf{p}\cdot\mathbf{r}\right). \qquad (10)
\end{aligned}
$$

让我们做一个有用的变换

$$
\left(y\frac{\partial}{\partial z} - z\frac{\partial}{\partial y}\right)\exp\left(\frac{\mathrm{i}}{\hbar}\mathbf{p}\cdot\mathbf{r}\right) \equiv \left(p_z\frac{\partial}{\partial p_y} - p_y\frac{\partial}{\partial p_z}\right)\exp\left(\frac{\mathrm{i}}{\hbar}\mathbf{p}\cdot\mathbf{r}\right). \qquad (11)
$$

不难验证

$$
\begin{aligned}
&\left(p_z\frac{\partial}{\partial p_y} - p_y\frac{\partial}{\partial p_z}\right)\exp\left(\frac{\mathrm{i}}{\hbar}\mathbf{p}\cdot\mathbf{r}\right) = p_z\frac{\partial}{\partial p_y}\exp\left(\frac{\mathrm{i}}{\hbar}\mathbf{p}\cdot\mathbf{r}\right) - p_y\frac{\partial}{\partial p_z}\exp\left(\frac{\mathrm{i}}{\hbar}\mathbf{p}\cdot\mathbf{r}\right) \\
&= p_z\frac{\partial}{\partial p_y}\exp\left(\frac{\mathrm{i}}{\hbar}(xp_x + yp_y + zp_z)\right) - p_y\frac{\partial}{\partial p_z}\exp\left(\frac{\mathrm{i}}{\hbar}(xp_x + yp_y + zp_z)\right) \\
&= p_z\frac{\mathrm{i}}{\hbar}y\exp\left(\frac{\mathrm{i}}{\hbar}(xp_x + yp_y + zp_z)\right) - p_y\frac{\mathrm{i}}{\hbar}z\exp\left(\frac{\mathrm{i}}{\hbar}(xp_x + yp_y + zp_z)\right)
\end{aligned}
$$

$$= \frac{\hbar}{\mathrm{i}}\frac{\partial}{\partial z}\frac{\mathrm{i}}{\hbar}y\exp\left(\frac{\mathrm{i}}{\hbar}(xp_x+yp_y+zp_z)\right) - \frac{\hbar}{\mathrm{i}}\frac{\partial}{\partial y}\frac{\mathrm{i}}{\hbar}z\exp\left(\frac{\mathrm{i}}{\hbar}(xp_x+yp_y+zp_z)\right)$$
$$= \left(y\frac{\partial}{\partial z} - z\frac{\partial}{\partial y}\right)\exp\left(\frac{\mathrm{i}}{\hbar}\mathbf{p}\cdot\mathbf{r}\right). \tag{12}$$

代入原式得

$$\begin{aligned}(\hat{L}_x)_{\mathbf{p}',\mathbf{p}} &= \int \mathrm{d}\mathbf{r}\frac{1}{(2\pi\hbar)^3}\exp\left(-\frac{\mathrm{i}}{\hbar}\mathbf{p}'\cdot\mathbf{r}\right)\frac{\hbar}{\mathrm{i}}\left(y\frac{\partial}{\partial z} - z\frac{\partial}{\partial y}\right)\exp\left(\frac{\mathrm{i}}{\hbar}\mathbf{p}\cdot\mathbf{r}\right)\\ &= \int \mathrm{d}\mathbf{r}\frac{1}{(2\pi\hbar)^3}\exp\left(-\frac{\mathrm{i}}{\hbar}\mathbf{p}'\cdot\mathbf{r}\right)\frac{\hbar}{\mathrm{i}}\left(p_z\frac{\partial}{\partial p_y} - p_y\frac{\partial}{\partial p_z}\right)\exp\left(\frac{\mathrm{i}}{\hbar}\mathbf{p}\cdot\mathbf{r}\right)\\ &= \int \mathrm{d}\mathbf{r}\frac{1}{(2\pi\hbar)^3}\exp\left(\frac{\mathrm{i}}{\hbar}(\mathbf{p}-\mathbf{p}')\cdot\mathbf{r}\right)\frac{\hbar}{\mathrm{i}}\left(p_z\frac{\partial}{\partial p_y} - p_y\frac{\partial}{\partial p_z}\right)\\ &= \frac{1}{(2\pi\hbar)^3}\int \exp\left(\frac{\mathrm{i}}{\hbar}(\mathbf{p}-\mathbf{p}')\cdot\mathbf{r}\right)\mathrm{d}\mathbf{r}\frac{\hbar}{\mathrm{i}}\left(p_z\frac{\partial}{\partial p_y} - p_y\frac{\partial}{\partial p_z}\right)\\ &= \frac{\hbar}{\mathrm{i}}\left(p_z\frac{\partial}{\partial p_y} - p_y\frac{\partial}{\partial p_z}\right)\delta(\mathbf{p}-\mathbf{p}').\end{aligned} \tag{13}$$

在上式的最后我们利用了 δ 函数的表达式

$$\frac{1}{(2\pi\hbar)^3}\int \exp\left(\frac{\mathrm{i}}{\hbar}(\mathbf{p}-\mathbf{p}')\cdot\mathbf{r}\right)\mathrm{d}\mathbf{r} = \delta(\mathbf{p}-\mathbf{p}'). \tag{14}$$

有关的详细介绍可参考曾谨言《量子力学卷 1》附录二.

2)

$$\begin{aligned}(\hat{L}_x^2)_{\mathbf{p}',\mathbf{p}} &= \langle\mathbf{p}'|\,\hat{L}_x^2\,|\mathbf{p}\rangle = \iint \mathrm{d}\mathbf{r}\mathrm{d}\mathbf{r}'\langle\mathbf{p}'\,|\mathbf{r}'\rangle\,\langle\mathbf{r}'|\,\hat{L}_x^2\,|\mathbf{r}\rangle\,\langle\mathbf{r}\,|\mathbf{p}\rangle\\ &= \iint \mathrm{d}\mathbf{r}\mathrm{d}\mathbf{r}'\left[\frac{1}{(2\pi\hbar)^{3/2}}\exp\left(\frac{\mathrm{i}}{\hbar}\mathbf{p}'\cdot\mathbf{r}'\right)\right]^* L_x^2\langle\mathbf{r}'\,|\mathbf{r}\rangle\,\frac{1}{(2\pi\hbar)^{3/2}}\exp\left(\frac{\mathrm{i}}{\hbar}\mathbf{p}\cdot\mathbf{r}\right)\\ &= \iint \mathrm{d}\mathbf{r}\mathrm{d}\mathbf{r}'\left[\frac{1}{(2\pi\hbar)^{3/2}}\exp\left(\frac{\mathrm{i}}{\hbar}\mathbf{p}'\cdot\mathbf{r}'\right)\right]^* L_x^2\delta(\mathbf{r}'-\mathbf{r})\frac{1}{(2\pi\hbar)^{3/2}}\exp\left(\frac{\mathrm{i}}{\hbar}\mathbf{p}\cdot\mathbf{r}\right)\\ &= \int \mathrm{d}\mathbf{r}\left[\frac{1}{(2\pi\hbar)^{3/2}}\exp\left(\frac{\mathrm{i}}{\hbar}\mathbf{p}'\cdot\mathbf{r}\right)\right]^* L_x^2\frac{1}{(2\pi\hbar)^{3/2}}\exp\left(\frac{\mathrm{i}}{\hbar}\mathbf{p}\cdot\mathbf{r}\right)\\ &= \int \mathrm{d}\mathbf{r}\frac{1}{(2\pi\hbar)^3}\exp\left(-\frac{\mathrm{i}}{\hbar}\mathbf{p}'\cdot\mathbf{r}\right)L_x^2\exp\left(\frac{\mathrm{i}}{\hbar}\mathbf{p}\cdot\mathbf{r}\right)\\ &= \int \mathrm{d}\mathbf{r}\frac{1}{(2\pi\hbar)^3}\exp\left(-\frac{\mathrm{i}}{\hbar}\mathbf{p}'\cdot\mathbf{r}\right)\left[-\hbar^2\left(y\frac{\partial}{\partial z} - z\frac{\partial}{\partial y}\right)^2\right]\exp\left(\frac{\mathrm{i}}{\hbar}\mathbf{p}\cdot\mathbf{r}\right).\end{aligned} \tag{15}$$

让我们做一个有用的变换

$$\left(y\frac{\partial}{\partial z} - z\frac{\partial}{\partial y}\right)^2\exp\left(\frac{\mathrm{i}}{\hbar}\mathbf{p}\cdot\mathbf{r}\right) \equiv \left(p_z\frac{\partial}{\partial p_y} - p_y\frac{\partial}{\partial p_z}\right)^2\exp\left(\frac{\mathrm{i}}{\hbar}\mathbf{p}\cdot\mathbf{r}\right). \tag{16}$$

不难验证

$$\left(p_z\frac{\partial}{\partial p_y} - p_y\frac{\partial}{\partial p_z}\right)^2\exp\left(\frac{\mathrm{i}}{\hbar}\mathbf{p}\cdot\mathbf{r}\right) = \left(p_z^2\frac{\partial^2}{\partial p_y^2} + p_y^2\frac{\partial^2}{\partial p_z^2} - 2p_yp_z\frac{\partial^2}{\partial p_y\partial p_z}\right)\exp\left(\frac{\mathrm{i}}{\hbar}\mathbf{p}\cdot\mathbf{r}\right)$$

$$
\begin{aligned}
&= p_z^2 \frac{\partial^2}{\partial p_y^2} \exp\left(\frac{\mathrm{i}}{\hbar}\mathbf{p}\cdot\mathbf{r}\right) + p_y^2 \frac{\partial^2}{\partial p_z^2} \exp\left(\frac{\mathrm{i}}{\hbar}\mathbf{p}\cdot\mathbf{r}\right) - 2p_y p_z \frac{\partial^2}{\partial p_y \partial p_z} \exp\left(\frac{\mathrm{i}}{\hbar}\mathbf{p}\cdot\mathbf{r}\right) \\
&= p_z^2 \frac{\partial^2}{\partial p_y^2} \exp\left(\frac{\mathrm{i}}{\hbar}(xp_x + yp_y + zp_z)\right) + p_y^2 \frac{\partial^2}{\partial p_z^2} \exp\left(\frac{\mathrm{i}}{\hbar}(xp_x + yp_y + zp_z)\right) \\
&- 2p_y p_z \frac{\partial^2}{\partial p_y \partial p_z} \exp\left(\frac{\mathrm{i}}{\hbar}(xp_x + yp_y + zp_z)\right). \qquad (17)
\end{aligned}
$$

将上式逐项求偏微商即可证得. 所以

$$
\begin{aligned}
(\hat{L}_x^2)_{\mathbf{p}',\mathbf{p}} &= \int \mathrm{d}\mathbf{r} \frac{1}{(2\pi\hbar)^3} \exp\left(-\frac{\mathrm{i}}{\hbar}\mathbf{p}'\cdot\mathbf{r}\right) \left[-\hbar^2\left(y\frac{\partial}{\partial z} - z\frac{\partial}{\partial y}\right)^2\right] \exp\left(\frac{\mathrm{i}}{\hbar}\mathbf{p}\cdot\mathbf{r}\right) \\
&= \int \mathrm{d}\mathbf{r} \frac{1}{(2\pi\hbar)^3} \exp\left(-\frac{\mathrm{i}}{\hbar}\mathbf{p}'\cdot\mathbf{r}\right) \left[-\hbar^2\left(p_z\frac{\partial}{\partial p_y} - p_y\frac{\partial}{\partial p_z}\right)^2\right] \exp\left(\frac{\mathrm{i}}{\hbar}\mathbf{p}\cdot\mathbf{r}\right) \\
&= -\hbar^2\left(p_z\frac{\partial}{\partial p_y} - p_y\frac{\partial}{\partial p_z}\right)^2 \delta(\mathbf{p}-\mathbf{p}'). \qquad (18)
\end{aligned}
$$

4.2

我们知道无限深一维方势阱中粒子在阱内 $0 \le x \le a$ 时, 其波函数为 $\psi(x) = \sqrt{\frac{2}{a}}\sin\left(\frac{n\pi x}{a}\right)$, 所以

1)

$$
\begin{aligned}
x_{n,m} &= \langle n|\,\hat{x}\,|m\rangle = \langle \psi_n|\,\hat{x}\,|\psi_m\rangle = \int \mathrm{d}\mathbf{r}\psi_n^*\hat{x}\psi_m \\
&= \int_0^a \mathrm{d}x\sqrt{\frac{2}{a}}\sin\left(\frac{n\pi x}{a}\right) x\sqrt{\frac{2}{a}}\sin\left(\frac{m\pi x}{a}\right) \\
&= \frac{2}{a}\int_0^a x\sin\left(\frac{n\pi x}{a}\right)\sin\left(\frac{m\pi x}{a}\right)\mathrm{d}x. \qquad (19)
\end{aligned}
$$

在此, 我们可以查阅积分表直接得到结果. 但是如果你热衷计算积分, 大可以想办法将上式手算出来. 我们下面就进行这个工作. 考虑到三角函数的积化和差公式

$$
\sin\alpha\sin\beta = -\frac{1}{2}[\cos(\alpha+\beta) - \cos(\alpha-\beta)]. \qquad (20)
$$

则 (19) 式化为

$$
\begin{aligned}
x_{n,m} &= \frac{2}{a}\int_0^a x\left(-\frac{1}{2}\left[\cos(\frac{n\pi x}{a} + \frac{m\pi x}{a}) - \cos(\frac{n\pi x}{a} - \frac{m\pi x}{a})\right]\right)\mathrm{d}x \\
&= \frac{2}{a}\int_0^a x\left(-\frac{1}{2}\left[\cos\frac{(n+m)\pi x}{a} - \cos\frac{(n-m)\pi x}{a}\right]\right)\mathrm{d}x \\
&= -\frac{1}{a}\int_0^a x\left[\cos\frac{(n+m)\pi x}{a} - \cos\frac{(n-m)\pi x}{a}\right]\mathrm{d}x \\
&= -\frac{1}{a}\int_0^a x\cos\frac{(n+m)\pi x}{a}\mathrm{d}x + \frac{1}{a}\int_0^a x\cos\frac{(n-m)\pi x}{a}\mathrm{d}x \\
&= -\frac{1}{a}\left[\left(\frac{a}{(n+m)\pi}x\sin\frac{(n+m)\pi x}{a}\right)_0^a - \int_0^a \frac{a}{(n+m)\pi}\sin\frac{(n+m)\pi x}{a}\mathrm{d}x\right]
\end{aligned}
$$

$$
\begin{aligned}
&+\frac{1}{a}\left[\left(\frac{a}{(n-m)\pi}x\sin\frac{(n-m)\pi x}{a}\right)_0^a-\int_0^a\frac{a}{(n-m)\pi}\sin\frac{(n-m)\pi x}{a}\mathrm{d}x\right]\\
&=-\frac{1}{a}\left[0-\int_0^a\frac{a}{(n+m)\pi}\sin\frac{(n+m)\pi x}{a}\mathrm{d}x\right]+\frac{1}{a}\left[0-\int_0^a\frac{a}{(n-m)\pi}\sin\frac{(n-m)\pi x}{a}\mathrm{d}x\right]\\
&=\frac{1}{(n+m)\pi}\int_0^a\sin\frac{(n+m)\pi x}{a}\mathrm{d}x-\frac{1}{(n-m)\pi}\int_0^a\sin\frac{(n-m)\pi x}{a}\mathrm{d}x\\
&=\frac{1}{(n+m)\pi}\left[-\frac{a}{(n+m)\pi}\cos\frac{(n+m)\pi x}{a}\right]_0^a-\frac{1}{(n-m)\pi}\left[-\frac{a}{(n-m)\pi}\cos\frac{(n-m)\pi x}{a}\right]_0^a\\
&=\frac{1}{(n+m)\pi}\left[-\frac{a}{(n+m)\pi}\cos(n+m)\pi+\frac{a}{(n+m)\pi}\right]\\
&-\frac{1}{(n-m)\pi}\left[-\frac{a}{(n-m)\pi}\cos(n-m)\pi+\frac{a}{(n-m)\pi}\right]\\
&=\frac{a}{(n+m)^2\pi^2}\left[1-(-1)^{n+m}\right]-\frac{a}{(n-m)^2\pi^2}\left[1-(-1)^{n-m}\right]\\
&=\frac{a}{\pi^2}\frac{1}{(n^2-m^2)^2}\left((n-m)^2\left[1-(-1)^{n+m}\right]-(n+m)^2\left[1-(-1)^{n-m}\right]\right)\\
&=\frac{a}{\pi^2}\frac{1}{(n^2-m^2)^2}\left[(n-m)^2-(n-m)^2(-1)^{n+m}-(n+m)^2+(n+m)^2(-1)^{n-m}\right]\\
&=\frac{a}{\pi^2}\frac{1}{(n^2-m^2)^2}[n^2+m^2-2mn-(n-m)^2(-1)^{n+m}\\
&-n^2-m^2-2mn+(n+m)^2(-1)^{n-m}]\\
&=\frac{a}{\pi^2}\frac{1}{(n^2-m^2)^2}[-4mn-(n-m)^2(-1)^{n+m}+(n+m)^2(-1)^{n-m}].
\end{aligned}\tag{21}
$$

由于 $n+m$ 与 $n-m$ 有相同的奇偶性，所以 $(-1)^{n+m}$ 与 $(-1)^{n-m}$ 有相同的取向，故

$$
\begin{aligned}
x_{n,m}&=\frac{a}{\pi^2}\frac{1}{(n^2-m^2)^2}[-4mn-(n-m)^2(-1)^{n-m}+(n+m)^2(-1)^{n-m}]\\
&=\frac{a}{\pi^2}\frac{1}{(n^2-m^2)^2}[-4mn+[(n+m)^2-(n-m)^2](-1)^{n-m}]\\
&=\frac{a}{\pi^2}\frac{1}{(n^2-m^2)^2}4mn[(-1)^{n-m}-1].
\end{aligned}\tag{22}
$$

显然，我们计算的前提是 $n\neq m$. 而当 $n=m$ 时，

$$
\begin{aligned}
x_{m,m}&=\langle m|\,\hat{x}\,|m\rangle=\langle\psi_m|\,\hat{x}\,|\psi_m\rangle=\int\mathrm{d}\mathbf{r}\psi_m^*\hat{x}\psi_m\\
&=\int_0^a\mathrm{d}x\sqrt{\frac{2}{a}}\sin\left(\frac{m\pi x}{a}\right)x\sqrt{\frac{2}{a}}\sin\left(\frac{m\pi x}{a}\right)\\
&=\frac{2}{a}\int_0^a x\sin\left(\frac{m\pi x}{a}\right)\sin\left(\frac{m\pi x}{a}\right)\mathrm{d}x.
\end{aligned}\tag{23}
$$

我们依然可通过查阅积分表或者仿照 (21) 式的方法计算，最后得到

$$
x_{m,m}=\frac{a}{2}.\tag{24}
$$

2)

$$
p_{n,m}=\langle n|\,\hat{x}\,|m\rangle=\langle\psi_n|\,\hat{p}\,|\psi_m\rangle=\int\mathrm{d}\mathbf{r}\psi_n^*\hat{x}\psi_m
$$

$$= \int_0^a \mathrm{d}x \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi x}{a}\right) \frac{\hbar}{\mathrm{i}} \frac{\partial}{\partial x} \sqrt{\frac{2}{a}} \sin\left(\frac{m\pi x}{a}\right)$$
$$= \frac{2\hbar m\pi}{\mathrm{i}^2 a} \int_0^a \sin\left(\frac{n\pi x}{a}\right) \cos\left(\frac{m\pi x}{a}\right) \mathrm{d}x. \tag{25}$$

利用三角函数的积化和差公式

$$\sin\alpha\cos\beta = \frac{1}{2}[\sin(\alpha+\beta) + \sin(\alpha-\beta)]. \tag{26}$$

代入 (25) 式，仿照前面的做法，容易得到当 $n \neq m$ 时

$$p_{n,m} = \frac{2\hbar m}{\mathrm{i}a} \frac{n}{n^2-m^2}[1-(-1)^{n-m}]. \tag{27}$$

而当 $n = m$ 时

$$p_{m,m} = 0. \tag{28}$$

4.3

一维谐振子在坐标表象中的能量本征函数为

$$\psi_n(x) = \left(\frac{\alpha}{\sqrt{\pi}2^n n!}\right)^{\frac{1}{2}} \exp\left(-\frac{1}{2}\alpha^2 x^2\right) \mathrm{H}_n(\alpha x), \qquad \alpha = \sqrt{\frac{m\omega}{\hbar}}. \tag{29}$$

其中 $\mathrm{H}_n(\alpha x)$ 为 Hermite 多项式. 利用表象变换

$$c_n(p) = \int_\Omega \psi_p^* \psi_n \mathrm{d}\mathbf{r} = \int \psi_p^*(x)\psi_n(x)\mathrm{d}x$$
$$= \int \frac{1}{\sqrt{2\pi\hbar}} \exp\left(-\frac{\mathrm{i}}{\hbar}px\right) \left(\frac{\alpha}{\sqrt{\pi}2^n n!}\right)^{\frac{1}{2}} \exp\left(-\frac{1}{2}\alpha^2x^2\right) \mathrm{H}_n(\alpha x)\mathrm{d}x. \tag{30}$$

即得动量表象中的能量本征函数 (未化简).

4.4

仿照 4.1 题的做法，不难得到

$$H_{p',p} = \langle p'|\hat{H}|p\rangle = \iint \mathrm{d}x\mathrm{d}x' \langle p'|x'\rangle \langle x'|\hat{H}|x\rangle \langle x|p\rangle$$
$$= \iint \mathrm{d}x\mathrm{d}x' \langle p'|x'\rangle \langle x'| -\frac{\hbar^2}{2m}\frac{\mathrm{d}^2}{\mathrm{d}x^2} + \frac{1}{2}m\omega^2x^2 |x\rangle \langle x|p\rangle$$
$$= \iint \mathrm{d}x\mathrm{d}x' \langle p'|x'\rangle \left[-\frac{\hbar^2}{2m}\frac{\mathrm{d}^2}{\mathrm{d}x^2} + \frac{1}{2}m\omega^2x^2\right] \langle x'|x\rangle \langle x|p\rangle$$
$$= \iint \mathrm{d}x\mathrm{d}x' \langle p'|x'\rangle \left[-\frac{\hbar^2}{2m}\frac{\mathrm{d}^2}{\mathrm{d}x^2} + \frac{1}{2}m\omega^2x^2\right] \delta(x-x')\langle x|p\rangle$$
$$= \int \mathrm{d}x \langle p'|x\rangle \left[-\frac{\hbar^2}{2m}\frac{\mathrm{d}^2}{\mathrm{d}x^2} + \frac{1}{2}m\omega^2x^2\right] \langle x|p\rangle$$
$$= \int \mathrm{d}x \frac{1}{\sqrt{2\pi\hbar}} \exp\left(-\frac{\mathrm{i}}{\hbar}p'x\right) \left[-\frac{\hbar^2}{2m}\frac{\mathrm{d}^2}{\mathrm{d}x^2} + \frac{1}{2}m\omega^2x^2\right] \frac{1}{\sqrt{2\pi\hbar}} \exp\left(\frac{\mathrm{i}}{\hbar}px\right)$$

$$= \int \mathrm{d}x \frac{1}{2\pi\hbar} \exp\left(-\frac{\mathrm{i}}{\hbar}p'x\right)\left[-\frac{\hbar^2}{2m}\frac{\mathrm{d}^2}{\mathrm{d}x^2}+\frac{1}{2}m\omega^2x^2\right]\exp\left(\frac{\mathrm{i}}{\hbar}px\right). \tag{31}$$

让我们做一个有用的变换

$$\left(-\frac{\hbar^2}{2m}\frac{\mathrm{d}^2}{\mathrm{d}x^2}+\frac{1}{2}m\omega^2x^2\right)\exp\left(\frac{\mathrm{i}}{\hbar}px\right) \equiv \left(\frac{p^2}{2m}-\frac{1}{2}m\omega^2\hbar^2\frac{\mathrm{d}^2}{\mathrm{d}p^2}\right)\exp\left(\frac{\mathrm{i}}{\hbar}px\right). \tag{32}$$

不难验证

$$\begin{aligned}
&\left(\frac{p^2}{2m}-\frac{1}{2}m\omega^2\hbar^2\frac{\mathrm{d}^2}{\mathrm{d}p^2}\right)\exp\left(\frac{\mathrm{i}}{\hbar}px\right) = \frac{p^2}{2m}\exp\left(\frac{\mathrm{i}}{\hbar}px\right)-\frac{1}{2}m\omega^2\hbar^2\frac{\mathrm{d}^2}{\mathrm{d}p^2}\exp\left(\frac{\mathrm{i}}{\hbar}px\right)\\
&=\frac{p^2}{2m}\exp\left(\frac{\mathrm{i}}{\hbar}px\right)-\frac{1}{2}m\omega^2\hbar^2\left(\frac{\mathrm{i}}{\hbar}x\right)\frac{\mathrm{d}}{\mathrm{d}p}\exp\left(\frac{\mathrm{i}}{\hbar}px\right)\\
&=\frac{p^2}{2m}\exp\left(\frac{\mathrm{i}}{\hbar}px\right)-\frac{1}{2}m\omega^2\hbar^2\left(\frac{\mathrm{i}}{\hbar}x\right)^2\exp\left(\frac{\mathrm{i}}{\hbar}px\right)\\
&=\frac{p^2}{2m}\exp\left(\frac{\mathrm{i}}{\hbar}px\right)+\frac{1}{2}m\omega^2x^2\exp\left(\frac{\mathrm{i}}{\hbar}px\right)=\left(-\frac{\hbar^2}{2m}\frac{\mathrm{d}^2}{\mathrm{d}x^2}+\frac{1}{2}m\omega^2x^2\right)\exp\left(\frac{\mathrm{i}}{\hbar}px\right).
\end{aligned} \tag{33}$$

代入原式得

$$\begin{aligned}
H_{p',p} &= \int \mathrm{d}x \frac{1}{2\pi\hbar}\exp\left(-\frac{\mathrm{i}}{\hbar}p'x\right)\left(-\frac{\hbar^2}{2m}\frac{\mathrm{d}^2}{\mathrm{d}x^2}+\frac{1}{2}m\omega^2x^2\right)\exp\left(\frac{\mathrm{i}}{\hbar}px\right)\\
&= \int \mathrm{d}x \frac{1}{2\pi\hbar}\exp\left(-\frac{\mathrm{i}}{\hbar}p'x\right)\left(\frac{p^2}{2m}-\frac{1}{2}m\omega^2\hbar^2\frac{\mathrm{d}^2}{\mathrm{d}p^2}\right)\exp\left(\frac{\mathrm{i}}{\hbar}px\right)\\
&= \left(\frac{p^2}{2m}-\frac{1}{2}m\omega^2\hbar^2\frac{\mathrm{d}^2}{\mathrm{d}p^2}\right)\frac{1}{2\pi\hbar}\int\exp\left(-\frac{\mathrm{i}}{\hbar}p'x\right)\exp\left(\frac{\mathrm{i}}{\hbar}px\right)\mathrm{d}x\\
&= \left(\frac{p^2}{2m}-\frac{1}{2}m\omega^2\hbar^2\frac{\mathrm{d}^2}{\mathrm{d}p^2}\right)\delta(p-p').
\end{aligned} \tag{34}$$

4.5

设

$$\Psi = \begin{pmatrix}\psi_1\\ \psi_2\\ \psi_3\end{pmatrix}. \tag{35}$$

为 $\hat{L^2}$ 和 $\hat{L_z}$ 共同表象中的矩阵.

我们先来看算符 $\hat{L_x}$ 在 $\hat{L^2}$ 和 $\hat{L_z}$ 共同表象中的本征值和归一化波函数. 此时的本征方程可被写为

$$\frac{\hbar}{\sqrt{2}}\begin{pmatrix}0&1&0\\1&0&1\\0&1&0\end{pmatrix}\Psi = \lambda\hbar\Psi. \tag{36}$$

或者

$$\frac{\hbar}{\sqrt{2}}\begin{pmatrix}0&1&0\\1&0&1\\0&1&0\end{pmatrix}\begin{pmatrix}\psi_1\\ \psi_2\\ \psi_3\end{pmatrix} = \lambda\hbar\begin{pmatrix}1&0&0\\0&1&0\\0&0&1\end{pmatrix}\begin{pmatrix}\psi_1\\ \psi_2\\ \psi_3\end{pmatrix}. \tag{37}$$

移项后可以得到如下的矩阵

$$\begin{pmatrix} -\lambda\hbar & \frac{\hbar}{\sqrt{2}} & 0 \\ \frac{\hbar}{\sqrt{2}} & -\lambda\hbar & \frac{\hbar}{\sqrt{2}} \\ 0 & \frac{\hbar}{\sqrt{2}} & -\lambda\hbar \end{pmatrix} \begin{pmatrix} \psi_1 \\ \psi_2 \\ \psi_3 \end{pmatrix} = 0. \tag{38}$$

若使上式有非零解，其系数矩阵的行列式必须为零，即

$$\det \begin{pmatrix} -\lambda\hbar & \frac{\hbar}{\sqrt{2}} & 0 \\ \frac{\hbar}{\sqrt{2}} & -\lambda\hbar & \frac{\hbar}{\sqrt{2}} \\ 0 & \frac{\hbar}{\sqrt{2}} & -\lambda\hbar \end{pmatrix} = \hbar \cdot \det \begin{pmatrix} -\lambda & \frac{1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{2}} & -\lambda & \frac{1}{\sqrt{2}} \\ 0 & \frac{1}{\sqrt{2}} & -\lambda \end{pmatrix} = 0. \tag{39}$$

不难解得

$$\lambda_1 = 0, \qquad \lambda_2 = 1, \qquad \lambda_3 = -1 \tag{40}$$

若取 $\lambda_1 = 0$，将之代入 (37) 式，有

$$\frac{\hbar}{\sqrt{2}} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} \psi_1 \\ \psi_2 \\ \psi_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}. \tag{41}$$

利用矩阵的乘法可得

$$\frac{\hbar}{\sqrt{2}} \begin{pmatrix} 0+\psi_2+0 \\ \psi_1+0+\psi_3 \\ 0+\psi_2+0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}. \tag{42}$$

即 $\psi_3 = -\psi_1, \psi_2 = 0$，那么未归一化的波函数可写为

$$\Psi_1 = \begin{pmatrix} \psi_1 \\ 0 \\ -\psi_1 \end{pmatrix}. \tag{43}$$

对此归一化可得

$$1 = \Psi_1^\dagger \Psi_1 = \begin{pmatrix} \psi_1^* & 0 & -\psi_1^* \end{pmatrix} \begin{pmatrix} \psi_1 \\ 0 \\ -\psi_1 \end{pmatrix} = 2\psi_1^2. \tag{44}$$

取 $\psi_1 = \frac{1}{\sqrt{2}}$，所以归一化的波函数

$$\Psi_1 = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ 0 \\ -\frac{1}{\sqrt{2}} \end{pmatrix}. \tag{45}$$

对应于 $\hat{L_x}$ 的本征值为 0.

若取 $\lambda_2 = 1$，将之代入 (37) 式，有

$$\frac{\hbar}{\sqrt{2}} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} \psi_1 \\ \psi_2 \\ \psi_3 \end{pmatrix} = \hbar \begin{pmatrix} \psi_1 \\ \psi_2 \\ \psi_3 \end{pmatrix}. \tag{46}$$

仿照上面的做法，利用矩阵的乘法易得

$$\begin{pmatrix} \frac{1}{\sqrt{2}}\psi_2 \\ \frac{1}{\sqrt{2}}(\psi_1+\psi_3) \\ \frac{1}{\sqrt{2}}\psi_2 \end{pmatrix} = \begin{pmatrix} \psi_1 \\ \psi_2 \\ \psi_3 \end{pmatrix}. \tag{47}$$

即 $\psi_2 = \sqrt{2}\psi_1, \psi_2 = \sqrt{2}\psi_3, \psi_1 = \psi_3$，那么未归一化的波函数可写为

$$\Psi_2 = \begin{pmatrix} \psi_1 \\ \sqrt{2}\psi_1 \\ \psi_1 \end{pmatrix}. \tag{48}$$

对此归一化可得

$$1 = \Psi_2^\dagger \Psi_2 = \begin{pmatrix} \psi_1^* & \sqrt{2}\psi_1^* & \psi_1^* \end{pmatrix} \begin{pmatrix} \psi_1 \\ \sqrt{2}\psi_1 \\ \psi_1 \end{pmatrix} = 4\psi_1^2. \tag{49}$$

取 $\psi_1 = \frac{1}{2}$，所以归一化的波函数

$$\Psi_2 = \begin{pmatrix} \frac{1}{2} \\ \frac{1}{\sqrt{2}} \\ \frac{1}{2} \end{pmatrix}. \tag{50}$$

对应于 $\hat{L_x}$ 的本征值为 $\hbar$.

同理，若取 $\lambda_3 = -1$，则易得最后的归一化的波函数

$$\Psi_3 = \begin{pmatrix} \frac{1}{2} \\ -\frac{1}{\sqrt{2}} \\ \frac{1}{2} \end{pmatrix}. \tag{51}$$

对应于 $\hat{L_x}$ 的本征值为 $-\hbar$.

综合以上三个讨论，我们可以得到从 $\hat{L^2}$ 和 $\hat{L_z}$ 共同表象变换到 L_x 表象的幺正变换矩阵

$$S = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{2} & \frac{1}{2} \\ 0 & \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{2} & \frac{1}{2} \end{pmatrix}. \tag{52}$$

所以对角化的矩阵 L_x 可被写为

$$\begin{aligned} L_x' = S^\dagger L_x S &= \begin{pmatrix} \frac{1}{\sqrt{2}} & 0 & -\frac{1}{\sqrt{2}} \\ \frac{1}{2} & \frac{1}{\sqrt{2}} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{\sqrt{2}} & \frac{1}{2} \end{pmatrix} \frac{\hbar}{\sqrt{2}} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{2} & \frac{1}{2} \\ 0 & \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{2} & \frac{1}{2} \end{pmatrix} \\ &= \begin{pmatrix} 0 & 0 & 0 \\ 0 & \hbar & 0 \\ 0 & 0 & -\hbar \end{pmatrix}. \end{aligned} \tag{53}$$

然后我们再考虑算符 $\hat{L}_y$ 在 $\hat{L}^2$ 和 $\hat{L}_z$ 共同表象中的本征值和归一化波函数. 仿照上面的做法，我们不难得到归一化了的波函数

$$\widetilde{\Psi}_1 = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ 0 \\ \frac{1}{\sqrt{2}} \end{pmatrix}. \tag{54}$$

对应于 $\hat{L}_y$ 的本征值为 0.

归一化了的波函数

$$\widetilde{\Psi}_2 = \begin{pmatrix} \frac{1}{2} \\ \frac{\mathrm{i}}{\sqrt{2}} \\ -\frac{1}{2} \end{pmatrix}. \tag{55}$$

对应于 $\hat{L}_y$ 的本征值为 $\hbar$.

归一化了的波函数

$$\widetilde{\Psi}_3 = \begin{pmatrix} \frac{1}{2} \\ -\frac{\mathrm{i}}{\sqrt{2}} \\ -\frac{1}{2} \end{pmatrix}. \tag{56}$$

对应于 $\hat{L}_y$ 的本征值为 $-\hbar$.

综合以上三个讨论，我们可以得到从 $\hat{L}^2$ 和 $\hat{L}_z$ 共同表象变换到 L_y 表象的幺正变换矩阵

$$\widetilde{S} = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{2} & \frac{1}{2} \\ 0 & \frac{\mathrm{i}}{\sqrt{2}} & -\frac{\mathrm{i}}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{2} & -\frac{1}{2} \end{pmatrix}. \tag{57}$$

所以对角化的矩阵 L_y 可被写为

$$L_y' = \widetilde{S}^\dagger L_y \widetilde{S} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & \hbar & 0 \\ 0 & 0 & -\hbar \end{pmatrix}. \tag{58}$$

4.6

连续性方程为

$$\frac{\partial}{\partial t} w(\mathbf{r}, t) + \nabla \cdot \mathbf{j}(\mathbf{r}, t) = 0. \tag{59}$$

其中 $w(\mathbf{r}, t)$ 为几率密度，$\mathbf{j}(\mathbf{r}, t)$ 为几率流密度.

值得注意的是，矩阵关系是算符关系的表现，几率密度 $w(\mathbf{r}, t)$ 是密度算符 $\rho = |\psi\rangle\langle\psi|$ 在坐标表象中的对角元，即 $w(\mathbf{r}, t) = \langle \mathbf{r}| \hat{\rho} |\mathbf{r}\rangle$. 完整的矩阵关系包含所有矩阵元，我们应该从密度算符出发进行研究. 密度算符随时间的变化可由 Schrödinger 方程来确定. 本题详细的解答可参考学习指导书 86 页.